

ERA Ecosystem

Virtual Server Hosting Requirements

Prepared for: Hostart LLC (Baku, Azerbaijan)

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Language: English | Purpose: technical quotation / RFP

1. Purpose

This document describes the technical requirements for hosting the ERA Ecosystem production platform on infrastructure provided by Hostart. We request a quotation for the configurations below. Pricing should not be constrained by a budget ceiling in this RFP - please quote the recommended and optional items separately.

2. Workload summary

ERA is a multi-application business platform (control plane + finance ERP + industry satellites) deployed as a Docker Compose stack on Linux.

- Target scale (Phase 1): up to 10,000 registered user accounts (not concurrent sessions).
- Expected concurrent sessions at launch: roughly 50-200; growth toward ~300-500.
- Approximate stack size: ~20 containers (PostgreSQL 16 with multiple databases, Redis 7, Traefik reverse proxy, NestJS APIs, multiple Next.js applications).
- Public entry: HTTPS via Traefik on ports 80/443; many subdomains on one public IPv4.
- Domain/DNS: remains under our control (era-365.online). Hostart provides server + public IP only.

3. Phase 1 - Primary production VM (all-in-one)

Initial deployment runs the full stack on a single virtual machine. Please quote the Target configuration as primary; Pilot may be offered as a lower-cost start option.

Parameter	Pilot (optional)	Target (preferred)
vCPU	8 dedicated	16 dedicated
RAM	32 GB	64 GB
Storage	200-320 GB NVMe SSD	400-500 GB NVMe SSD
Network	1 Gbps uplink	1 Gbps uplink
Transfer	Unmetered or >= 5 TB/mo	Unmetered or >= 7 TB/mo
Virtualization	KVM (not OpenVZ)	KVM (not OpenVZ)
Public IPv4	1 address	1 address
IPv6	Optional	Optional

CPU must be dedicated (or clearly guaranteed). Shared/burstable-only plans are not suitable for this workload. NVMe SSD is required; HDD is not acceptable for the primary disk.

4. Phase 2 - Database separation (required growth path)

A single VM is acceptable for launch, but we consider separating PostgreSQL (and preferably Redis) onto dedicated infrastructure as a necessary next step for stability under growth. Please quote Phase 2 as an optional add-on / upgrade path (not mandatory on day one).

4.1 Preferred Phase 2 layout

- VM A - Application tier: Docker Compose apps + Traefik (e.g. 8-16 vCPU / 32-64 GB RAM / 200+ GB NVMe).

- VM B - Data tier: PostgreSQL 16 (+ optional Redis), private network only for DB ports (e.g. 4-8 vCPU / 32-64 GB RAM / 400+ GB NVMe, backups attached).
- Private networking between VMs (VLAN / VPC / private IP) is required for Phase 2.
- Please confirm whether vertical resize (CPU/RAM) and adding a second VM are supported without full rebuild.

5. Optional - Staging environment

Please quote an optional smaller second VM for staging/UAT (same OS and Docker capability). Suggested starting size: 4-8 vCPU / 16-32 GB RAM / 150-200 GB NVMe. Staging may share the same firewall patterns; it must have a separate public IP or clear routing plan.

6. Operating system and software requirements

- OS: Ubuntu 24.04 LTS (x86_64) preferred; Ubuntu 22.04 LTS acceptable.
- Access: root or sudo-capable SSH user; SSH key authentication; password login disabled preferred.
- We prefer unmanaged / self-managed administration. Optional managed OS patching may be quoted separately.
- Docker Engine 24+ and Docker Compose v2 must be allowed (we can install; please confirm no policy blocks containers).
- cPanel / shared hosting control panels are not required and must not restrict Docker.
- Outbound HTTPS (443) and package repositories must be reachable for updates and image pulls.
- Ability to run 20+ concurrent containers without artificial process/container caps.
- Reasonable ulimits / file descriptor limits suitable for Node.js and PostgreSQL.

7. Network, firewall, and DNS

7.1 Ports

- MUST be reachable from Internet: TCP 22 (SSH), 80 (HTTP), 443 (HTTPS).
- MUST NOT be exposed to Internet: PostgreSQL 5432, Redis 6379 (and similar data ports).
- Cloud or host firewall configuration support is required (UFW and/or provider firewall).

7.2 DNS (our side)

- We manage DNS for era-365.online (and related zones).
- We will create multiple A records pointing to the provided public IP (app, api, finance-core, hotel-pms, etc.).
- Please provide the public IPv4 promptly after provisioning; reverse DNS (PTR) is a plus if available.

8. Security, backups, and operations

- Data center location (MANDATORY): all compute and primary data storage must be physically located in Azerbaijan (Baku preferred). Please confirm the facility name in writing.
- DDoS protection: please describe what is included at network/DC level.
- Snapshots: VM snapshot capability and frequency options.
- Backups: please describe available backup products (disk image and/or filesystem). We will also run application-level PostgreSQL dumps; off-server backup storage options are of interest (backup copies must also remain in Azerbaijan unless otherwise agreed in writing).
- SLA: please state monthly uptime SLA and maintenance window policy.
- Support: channels, languages (AZ/RU/EN), business-hours vs 24/7, response time targets.
- NDA (MANDATORY): a mutual Non-Disclosure Agreement must be executed before we share detailed architecture, credentials, or production access. Please send your standard NDA or confirm you will sign ours.

9. Data residency and confidentiality (mandatory)

The following are hard requirements for this engagement:

- Data residency: production workloads and primary databases must reside on infrastructure physically located in the Republic of Azerbaijan. Cross-border replication or storage outside Azerbaijan is not permitted without prior written approval.
- Confidentiality: mutual NDA is required prior to exchanging non-public system details.

Out of scope for this initial quotation (do not block the quote on these): banking-grade / CBAR-specific compliance certifications, formal audit attestations, and sector-specific regulatory addenda. These may be discussed later if needed.

10. Questions for the future (not required now - please comment if available)

- Managed PostgreSQL or dedicated database appliance offerings?
- Managed Redis?
- Object storage (S3-compatible) in the same DC for backups and file uploads?
- Kubernetes / container platform (not needed for Phase 1)?
- Multi-AZ or secondary site for disaster recovery?
- Floating IP / failover IP?
- Can we scale to significantly higher concurrent sessions later on your platform?

11. Deliverables we need in your quotation

- Monthly and setup fees for: Pilot VM (optional), Target VM, Phase 2 data VM, Staging VM (optional).
- Included traffic, IPv4, firewall, snapshots, backups, DDoS - itemized.
- Provisioning lead time.
- Resize policy (CPU/RAM/disk) and any downtime expectations.
- Confirmation that Docker Compose production workloads are supported.
- Written confirmation that all production compute and primary data storage are physically in Azerbaijan (facility name).
- NDA: your template or confirmation that you will sign ours (required before detailed handoff).
- Data center name / address class (city) and uptime SLA.

12. Contact / next step

Please reply with a written quotation matching sections 3-5 and answers to sections 8 and 10. We can schedule a short technical call after the quote.

Document generated for vendor evaluation. Application architecture details beyond this scope can be shared under NDA.